

1(a)	Gravitational potential energy of a mass at a point is defined as the work done on the mass in moving it from infinity to that point.
(b)(i)	Change in gravitational potential $\Delta\phi = -\frac{GM}{3R} - \left(-\frac{GM}{2R}\right) = \frac{GM}{6R}$ Hence the change in gravitational potential energy $\Delta U = m \Delta\phi = \frac{GMm}{6R}$
(b)(ii)	Gain in gravitational potential energy = Loss in kinetic energy $\Delta U = \frac{1}{2} mu^2 - \frac{1}{2} mv^2$ $\frac{GMm}{6R} = \frac{1}{2} m (7600^2 - 7320^2)$ $\frac{GM}{6R} = (2.09 \times 10^6)$ $(6.67 \times 10^{-11} M) / (6 \times 3.4 \times 10^6) = 2.09 \times 10^6$ $M = 6.39 \times 10^{23} \text{ kg}$
(b)(iii)	Ignore friction with atmosphere / Rocket is outside atmosphere / No energy is lost as heat and sound Not influenced by another planet. Mass of rocket is assumed constant / mass of fuel is negligible. Any two.
(c)(i)	The gravitational force acting on the satellite provides the centripetal force for the circular motion. Since gravitational force acts towards the centre of the earth, the circular orbit must be centred on the Earth's centre. If the orbit is not vertically above the equator, the satellite would have varying latitude and so cannot be geostationary.
(c)(ii)	$mr\omega^2 = \frac{GMm}{r^2}$ $r^3 = \frac{GM}{\omega^2} = \frac{GMT^2}{4\pi^2}$ $= \frac{6.67 \times 10^{-11} \times (24 \times 60 \times 60)^2 M}{4\pi^2}$ $r = 0.233 M^{1/3}$

2 (a) the energy per unit mass required to change a substance from the liquid phase to the